1. A sound wave is produced when an object ...

A. accelerates	
B. decelerates	
C. vibrates	
D. remains stationary	

2. The wavelength of a wave is measured in ...

A. metres

B. hertz

C. seconds

D. decibels

3. An echo occurs when a sound wave is ...

A. absorbed

B. transmitted	
C. refracted	
D. reflected	

4. Another name for the unit hertz is …

A. cycles per second	
B. seconds per cycle	

C. metres per second

D. decibels

5. Sound travels fastest in ...

A. a vacuum

B. the sea

C. the atmosphere

D. a broom stick

6. Which of the following does **not** describe a sound wave?

A. transverse wave

B. longitudinal wave

C. compression wave

D. push-pull wave

7. An animal that can hear sound frequencies higher than a human child is ...

A. a human adult	
B. a cod fish	
C. a bat	
D. an eagle	

8. Hitting a drum harder makes the sound …

A. higher	
B. lower	
C. louder	
D. softer	

9. The part of the ear that responds to sound waves like a microphone’s diaphragm is the …

A. lobe	

B. eardrum

C. bones of the middle ear

D. fluid in the inner ear

10. Sounds above 20000Hz are called …

A. ultracool

B. ultrasound

C. infra–audio	
D. infrasound	

Revision questions – sound waves

1. What type of movement sets up a sound wave?

2. In what units are the following measured?

   (a) frequency (b) wavelength (c) loudness

3. If a tuning fork has a frequency of 250Hz, how many cycles per second does it vibrate?

4. Why does sound travel faster in a solid than in a gas or liquid?

5. Describe a longitudinal wave.

6. Describe a transverse wave.

7. What type of wave is a sound wave?

8. What happens when a sound wave hits the eardrum?

9. Who is most likely to hear a bat; an adult or a child?

10. Draw an oscilloscope trace to show a note starting low and getting higher.

11. Draw an oscilloscope trace to show a note starting soft and getting louder.

12. How can you make a drum sound louder

13. How can you make a guitar string sound higher? (2 methods)

14. What is meant by ultrasound?

15. Why should we not stand too close to the band at an outdoor event?

Teaching notes and answers

There is a PowerPoint version of the multiple choice quiz which would make a useful starter or plenary activity.

The revision questions could be used in a variety of ways such as;

- A straightforward test, quiz or homework activity
- Give each student an answer or a question and get them to find their partner
- Use with one of our revision games such as ‘Science Ludo’ (22540) or ‘Science challenge track’ (22541).

Multiple choice quiz answers

1C; 2A; 3 D; 4A; 5D; 6A; 7C; 8C; 9B; 10B

Revision questions answers

1. A vibration
2. a) hertz b) metres c) decibels
3. 250
4. Because particles are closer together in a solid

5. This is a push-pull wave
6. This is a shake wave
7. Longitudinal
8. The eardrum vibrates
9. A child
10. Drawing showing bunching of waves increasing; wavelength decreasing
11. Drawing of wave getting taller; increasing in amplitude
12. Hit it harder
13. Tighten the string; shorten the string
14. Sound above 2000Hz; sound above human hearing
15. Damage to our hearing